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English-Normalized Text and Topic Analytics for FixMyStreet Brussels: Spatio-Temporal Hotspot Detection and Decision Support from Citizen Reports

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Abstract

Citizen-reporting platforms generate high-volume, multilingual streams of service requests, yet operational triage often relies on coarse category labels and manual inspection. This study develops an explainable analytics pipeline with probability calibration for FixMyStreet Brussels reports, combining text-based urgency modeling, topic discovery, and spatio-temporal hotspot scoring to inform municipal analytic review. From 522,132 raw reports, we build an English-normalized text field for modeling, derive resolution-time outcomes from closed cases, and curate a 1000-item gold standard with an explicit high-urgency class. A TF-IDF logistic regression baseline achieves reasonable classification performance on the labeled split and, after probability calibration, yields confidence estimates that are more suitable for risk-aware prioritization than uncalibrated scores. Topic-level analyses reveal dominant themes related to sidewalks, road damage, and bulky waste, and hotspot scores highlight persistent, high-impact issue clusters. Event detection on aggregated signals did not identify events above the predefined z-score threshold during the analysis window, suggesting that the observed dynamics are more visible as chronic, recurring problems than as abrupt threshold-level anomalies. Explainability audits via Shapley Additive Explanations (SHAP) expose linguistically intuitive drivers for urgent cases (e.g., dangerous, risk, and accident) and complaint-oriented terms (e.g., abandoned, illegal, and dirty), providing transparent hooks for governance review. The analysis is therefore presented as an open-data, English-normalized decision-support prototype rather than as a validated native multilingual triage system. The labeled evidence base contains 2200 distinct human-reviewed reports. It comprises the 1000-report gold standard, a 200-report model-ranked high-urgency candidate set, and a 1000-report expanded validation subset. The expanded validation subset contains 439 high-urgency cases, including 42 cases from a random 500-report corpus sample. To avoid overstating language evidence, this study makes no native multilingual claim. The empirical claim is limited to English-normalized text analytics over reports that were originally submitted in a multilingual civic-reporting setting.

Keywords: citizen sensing; FixMyStreet; English-normalized civic text analytics; urgency classification; topic modeling; hotspot detection; calibration; explainable AI

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1. Introduction

Digital civic-participation tools have expanded the role of citizens in urban service monitoring, enabling residents to report local problems such as illegal dumping, damaged sidewalks, or lighting failures at fine spatial and temporal granularity. These reports

constitute a form of volunteered geographic information and can complement official sensing and inspection regimes by increasing coverage and responsiveness [1,2]. At the same time, high submission volumes and heterogeneous free text create operational bottlenecks: municipal teams must triage competing requests, manage backlogs, and identify emerging trouble spots while maintaining accountability. Decision-support methods that integrate language signals with spatio-temporal patterns are therefore increasingly relevant for public administrations seeking data-driven service allocation.

FixMyStreet platforms provide a useful testbed for such methods because they mix structured fields (category, status, timestamps, and location) with citizen narratives that often encode urgency cues and situational context. However, text inputs are multilingual, noisy, and frequently short; labels may reflect administrative routing rather than risk; and predicted "priority" scores must be interpretable and properly calibrated to be actionable. The goal of this work is to deliver a transparent analytics pipeline that (i) models report urgency from text in a way that supports risk-aware ranking, (ii) summarizes recurrent problem themes via topic models, and (iii) quantifies spatio-temporal hotspots and potential event-like deviations using simple, auditable statistics.

The contribution is therefore not a new classifier or a deployed triage system. It is an open-data operational audit design that joins calibrated urgency ranking, rare-class second-human validation, topic and hotspot views, and a metadata-only circularity check for one large civic-reporting platform.

2. Literature Review

Research on citizen reporting and urban informatics has emphasized the value of crowdsourced observations for municipal services, planning, and crisis response [1]. Empirical work on volunteered geographic information has also highlighted systematic quality challenges—uneven participation, positional uncertainty, and reporting bias—which motivate methods that combine human oversight with automated screening [2]. Research on Open311 and 311 systems also shows that citizen-reporting platforms make infrastructure more visible to both residents and government, but the reports are shaped by reporting channels, neighborhood context, and participation behavior [3–7].

Natural language processing has been widely used to extract sentiment, complaints, and urgency from short texts, including social media and customer-feedback streams. In public-service contexts, urgency signals are often expressed through hazard and safety language (e.g., danger and accident) and through negative affect related to persistence or recurrence (e.g., again and always). Transformer architectures such as BERT have improved semantic representations for downstream classification tasks [8], and multilingual pretraining (e.g., XLM-R) supports crosslingual transfer where labeled data are sparse [9]. Nevertheless, linear baselines remain attractive in governance settings due to their transparency and ease of audit, especially when paired with post hoc explanations.

Topic modeling is commonly used to organize large text corpora into interpretable themes. Classical probabilistic approaches such as latent Dirichlet allocation (LDA) provide a principled generative framework [10], while newer neural topic models combine contextual embeddings with class-based term weighting to improve coherence on short texts [11]. For municipal operations, topic models can surface recurring issue types, connect them to administrative categories, and facilitate targeted interventions.

Spatial and spatio-temporal hotspot analysis has a long history in geography and epidemiology. Local indicators of spatial association and scan statistics are frequently used to detect clusters and prioritize field action [12,13]. When operational decision making requires transparent heuristics, z-score-based anomaly indices and simple aggregation rules can offer useful, interpretable signals—particularly when combined with language-derived

severity probabilities. Event detection in time series has also been extensively studied, including change-point methods that identify distributional shifts or mean changes [14].

Finally, deploying predictive models in public administration requires attention to reliability and accountability. Probability calibration is essential when model scores are used as risk estimates rather than as mere rankings. Miscalibrated models can mislead triage, either by overstating certainty or by underestimating rare but critical risks. Post hoc calibration methods and reliability diagrams are standard tools for evaluating probabilistic correctness [15,16]. Explainability methods such as SHAP provide feature-level attributions compatible with linear models and can support governance audits [17].

3. Materials and Methods

The analysis uses an export of FixMyStreet Brussels reports as seen in Table 1, containing identifiers, timestamps, hierarchical categories, responsible organizational units, status, address descriptors, free-text narratives, and point geometries. Resolution time is computed from created and closed timestamps for closed cases.

Table 1. Data schema overview for FixMyStreet Brussels reports.

Field(s)	Type	Description
FID, gid, fims_id	Identifier	Unique record identifiers for traceability.
createddate, updateddate, closeddate	Temporal	Lifecycle timestamps for calculating resolution time and age.
category, head_category	Taxonomy	Hierarchical issue classification (e.g., Public Cleanliness).
responsible_org, responsible_dep	Administrative	Department responsible for intervention.
status	Operational	Current state (e.g., Open, Closed, and Transferred).
road_fr, road_nl, pccp	Location	Address descriptors and postal codes.
comment, comment_reporter	Narrative	Free-text citizen report and source indicator.
comment_translated	Derived text	English-normalized text used for modeling.
geom	Spatial	Point geometry (EPSG:4326/31370).

Free-text comments are cleaned by removing obvious system templates and non-informative tokens, standardizing whitespace and line breaks, and filtering empty entries. A language-normalization step produces an English field (comment_translated) used for downstream modeling. Short texts (<10 characters) are retained but flagged for robustness checks, since they provide limited semantic evidence. The gold-standard annotation scheme uses a stratified sample of 1000 reports manually labeled into three classes representing non-urgent/neutral (class 0), routine complaint/service request (class 1), and high urgency/safety risk (class 2). The high-urgency class is intentionally rare to reflect operational reality and to stress-test recall and calibration for critical cases (see Table 2). Because this study relies only on public FixMyStreet Brussels open data, municipal dispatch logs, protected attributes, socioeconomic indicators, and non-public workflow metadata were not available. To strengthen label validation within these constraints, a second human annotator independently relabeled all 1000 gold-standard reports from a blind text-only file, without access to original labels, model scores, categories, departments, status, postal codes, or resolution-time fields. Agreement between the original labels and the second annotation was then assessed with percent agreement, Cohen's kappa, quadratic weighted kappa, class-specific agreement, and a confusion matrix. The same annotator also reviewed 200 model-ranked high-urgency candidate reports from the existing corpus; these cases were used as a selective rare-class validation set, not as an unbiased test set. The use of percent agreement, kappa, weighted kappa, and class-specific agreement follows standard recommendations for computational-linguistic annotation reliability, where agreement coefficients must be interpreted together with task ambiguity and class prevalence [18].

Table 2. Preprocessing and sampling statistics.

Metric	Value	Notes
Total raw reports	522,132	Initial ingestion
Exact duplicates	760	Collapsed based on timestamp + geometry + text
Closed reports	426,490	Used for resolution-time calculation
Empty/null comments	14.25%	Excluded from text modeling
Short text (<10 chars)	15.57%	Flagged for robustness checks
Gold-standard size	1000	Manually annotated subset
Gold: high urgency (class 2)	50	Safety risks/urgent items

Table 3 reports the label-reliability audit for the gold-standard annotation, including the blind second annotation, agreement coefficients, class-specific agreement, and the selective high-urgency candidate review.

Table 3. Label reliability audit for the gold-standard annotation.

Validation Item	Value	Interpretation
Blind second annotation	1000 gold-standard reports	The second annotator received text only. Original labels, model scores, categories, departments, status, postal codes, and resolution-time fields were hidden.
Overall agreement	0.531	Label reliability was measured in the revised manuscript, not left unreported.
Cohen's kappa	0.246	Overall agreement was modest because many neutral reports were relabeled as routine complaints.
Quadratic weighted kappa	0.331	Ordinal disagreement remained limited enough to separate high-urgency from routine cases more clearly than neutral from routine cases.
Original class 0 agreement	170/605	The neutral/routine boundary was unstable.
Original class 1 agreement	322/345	Routine complaint labels were comparatively stable.
Original class 2 agreement	39/50	The high-urgency boundary had stronger support than the neutral boundary.
Additional candidate review	200 model-ranked high-urgency reports, 200/200 labeled class 2	This supports rare-class enrichment. It is not an unbiased performance estimate because the cases were model-selected.

The label reliability audit is reported before model performance so that classification results are interpreted against the stability of the human labels.

The validation design reduces dependence on the original small gold set through a larger blind text-only annotation subset. It uses the same second human annotator and includes 500 random corpus reports, 300 model-ranked high-urgency candidate reports, and 200 borderline or uncertain candidate reports. The annotation file excludes model scores, original labels, categories, departments, status fields, postal codes, and resolution-time outcomes. Table 4 summarizes the expanded validation subset design and separates the random corpus stratum from the rare-class and boundary-focused strata.

Table 4. Expanded validation subset design. The table separates the random corpus stratum from rare-class and boundary-focused strata.

Annotation Stratum	N	Selection Purpose	Use in Interpretation
Random corpus reports	500	Estimate label distribution and model behavior on new non-enriched cases.	Best new evidence for generalization, but still one annotator.
Model-ranked high-urgency candidates	300	Increase rare high-urgency evidence beyond the original 50 gold cases and the earlier 200 candidates.	Rare-class enrichment. Not prevalence-representative.
Borderline or uncertain candidates	200	Stress-test the routine versus high-urgency boundary.	Boundary analysis. Not prevalence-representative.
Expanded validation subset	1000	Contains distinct human-labeled reports.	Used as an expanded validation subset, not as a municipal deployment test.

The random stratum is the most relevant part for generalization. The model-ranked and borderline strata deliberately increase the number of difficult or high-risk cases, so they are interpreted as validation stress tests rather than prevalence estimates.

Together with the original 1000-report gold standard and the earlier 200 high-urgency candidate review, the distinct human-reviewed evidence base now covers 2200 reports. The design still uses one second annotator and therefore does not replace adjudicated multi-annotator labeling.

A bag-of-words baseline is trained using TF-IDF features and multinomial logistic regression. Model quality is assessed with accuracy, macro-averaged F1, class-specific recall, ranking-oriented metrics, and bootstrap uncertainty intervals. Because downstream decisions require interpretable probabilities, model confidence is evaluated via reliability diagrams and expected calibration error (ECE). A calibrated variant is obtained by fitting a post hoc calibration model on held-out predictions. Confusion matrices support error analysis across classes.

The implementation uses TF-IDF features with lowercase normalization, Unicode accent stripping, sublinear term frequency, max_df = 0.95, min_df = 5, and the unigram setting selected by five-fold stratified grid search. Logistic regression is fitted with C = 2.0 and balanced class weights. The best cross-validated macro-F1 during grid search was 0.8053. Probability calibration uses sigmoid calibration with three internal folds, while the fixed test split is retained for comparability with the confusion matrices.

Supplementary transformer baselines are evaluated without new labels by extracting frozen mean-pooled embeddings from distilbert-base-multilingual-cased and xlm-roberta-base and then fitting the same balanced logistic regression classifier in five-fold stratified cross-validation. These checks compare representational alternatives under the same small gold-standard constraint; they are not full fine-tuned transformer classifiers.

To summarize recurrent problem types, reports are clustered into topics using a term-based topic representation that yields top words per topic. For each topic, the mean predicted urgency probability and an aggregate hotspot score are computed, enabling comparisons between high-volume but low-risk themes and lower-volume themes that concentrate risk.

Topic modeling is implemented as non-negative matrix factorization (NMF) over TF-IDF features, with 20 topics, 15 top words per topic, max_features = 20,000, min_df = 5, max_df = 0.95, and n-grams up to bigrams. Supplementary checks compare NMF with latent Dirichlet allocation (LDA) on a 50,000-report sample using UMass coherence and BERTopic

on a 10,000-report sample using paraphrase-multilingual-MiniLM-L12-v2 embeddings, UMAP reduction, and HDBSCAN clustering. These comparisons are included because short administrative texts can produce unstable or artifact-heavy topics [11,19].

Reports are aggregated over spatial units and weekly time bins. Hotspot scores combine volume and urgency signals to highlight areas with persistent, high-impact demand. A z-score-based event detector is applied to aggregated time series; events are flagged only when deviations exceed a predefined threshold, reducing sensitivity to routine seasonality and noise.

For each area-week cell, the hotspot score is calculated as log(1 + volume) × (0.2 + mean predicted high-urgency probability). The event detector applies rolling z-scores to weekly volume, mean high-urgency probability, and hotspot score, using a six-week rolling window, minimum volume of three reports, and a z-threshold of 2.5. Sensitivity checks vary the z-threshold from 1.5 to 3.0 and the rolling window from 4 to 12 weeks, consistent with the broader change-point literature that treats anomaly detection as sensitive to temporal scale and thresholding choices [20].

For the TF–IDF logistic regression model, SHAP values are computed to identify terms that increase or decrease the predicted probability of high urgency and complaint-oriented classes. Term rankings are inspected for face validity and for potential governance concerns (e.g., artifacts and overly generic tokens).

Because this is a public-service context, model outputs are treated as reviewable decision-support signals rather than automated decisions. This framing aligns with recent AI risk-management guidance that emphasizes mapping, measuring, monitoring, transparency, and governance controls for AI systems used in consequential settings [21,22]. No municipal deployment, staff user study, or direct comparison with internal triage workflows was available within the open-data design. The operational checks are therefore retrospective and decision-oriented, using ranking metrics, priority-decile resolution patterns, and area-week summaries as indirect evidence rather than as proof of workflow improvement.

4. Results

As seen in Table 2, the export contained 522,132 raw reports. Exact duplicates were rare (760 records) and were collapsed using timestamp, geometry, and text keys. Closed reports represented the majority of cases (426,490) and were used to compute resolution-time outcomes. Text fields were frequently sparse: a total of 14.25% of reports had empty or null comments and were excluded from text modeling, and 15.57% contained fewer than 10 characters and were retained as low-information cases.

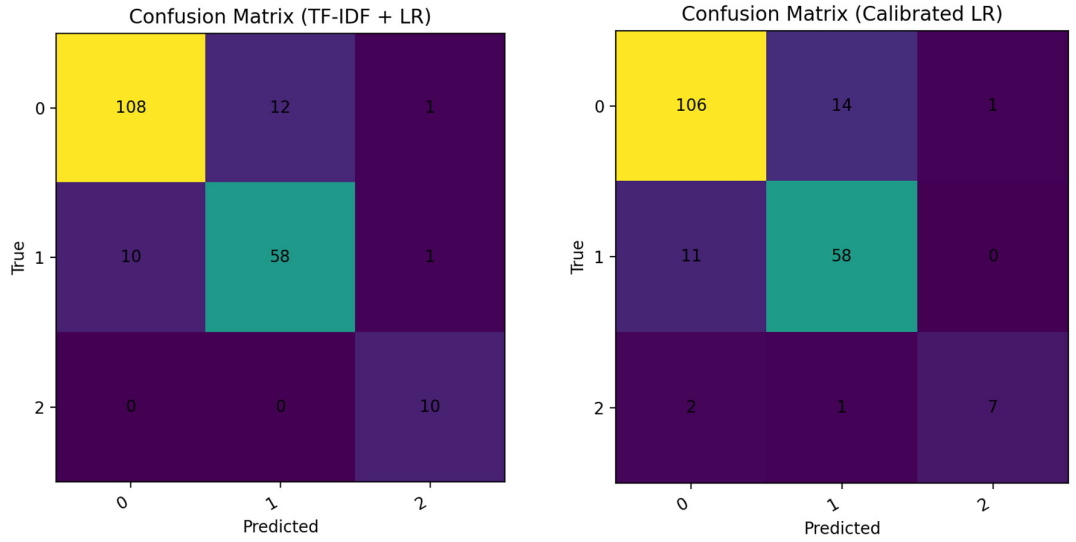
Table 5 shows that the calibrated TF-IDF logistic regression model reached an accuracy of 0.855 and a macro-F1 of 0.826 on the gold-standard test set, with a high-urgency class recall of 0.700. The uncalibrated model variant produced higher overall scores on this split, with an accuracy of 0.880 and a macro-F1 of 0.882, but it showed poorer probability behavior in reliability analysis.

Table 5. Urgency classification performance on the gold-standard split. The ECE for the calibrated model is 0.069.

Model	Accuracy	Macro-F1	Urgency (Class 2) Recall
TF–IDF + LR (raw)	0.880	0.882	1.000
TF–IDF + LR (calibrated)	0.885	0.826	0.700

Confusion matrices (Figure 1) indicate that most errors occur between classes 0 and 1, consistent with overlap between neutral reporting and routine complaints. High-urgency

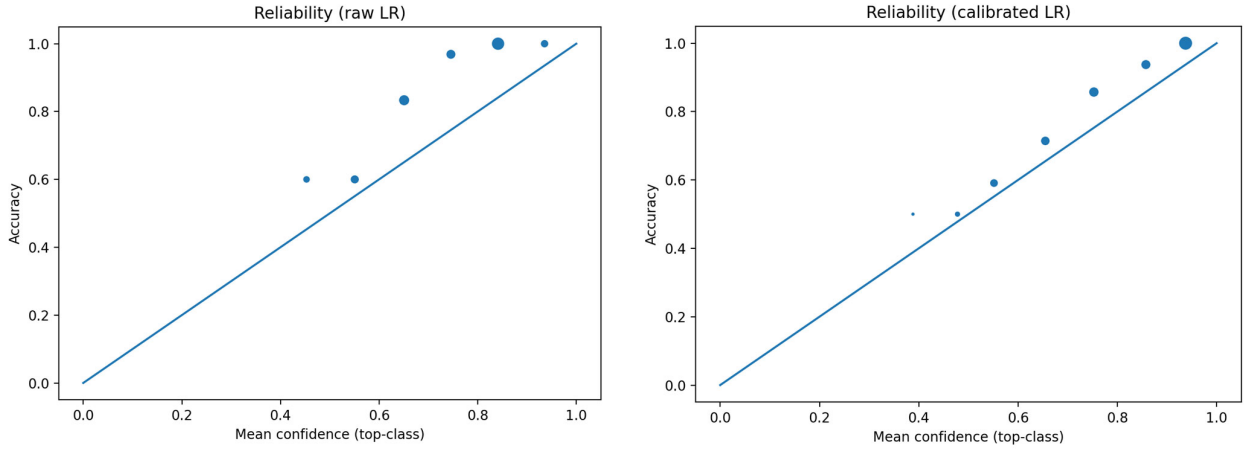
cases are typically recognized, but calibration shifts probability mass toward more conservative predictions, increasing some misses of class 2 in exchange for improved probability trustworthiness.



(**a**) (**b**)

Figure 1. Confusion matrix: (**a**) Raw TF–IDF logistic regression urgency classifier (3 classes). Values are counts on the gold-standard test split. (**b**) Calibrated TF–IDF logistic regression urgency classifier. Calibration yields more conservative class 2 predictions on the same evaluation split. Cell colors indicate count magnitude, with darker colors representing lower counts and brighter colors representing higher counts.

Reliability diagrams show that the calibrated model's mean confidence aligns more closely with empirical accuracy across bins, while the raw model tends to be overconfident at higher predicted probabilities, as seen in Figure 2. The calibrated model achieved an ECE of 0.069, supporting the use of its probabilities as decision-support scores rather than only as rankings.



(**a**) (**b**)

Figure 2. Reliability diagram: (**a**) Raw TF–IDF logistic regression model; (**b**) Calibrated TF–IDF logistic regression model. The diagonal line represents perfect calibration, where mean predicted confidence equals empirical accuracy. The dots represent calibration bins. Dots below the diagonal indicate overconfident predictions, while dots above the diagonal indicate underconfident predictions. Calibration improves the alignment between mean confidence and empirical accuracy..

Additional evaluation on the same labeled test split used ranking-oriented metrics for class 2. For the calibrated model, the top 10 reports ranked by predicted high-urgency probability contained 9 of the 10 high-urgency cases (top-10 recall = 0.900; precision at 10 = 0.900), and the top 20 contained all 10 cases. Average precision for class 2 was 0.957 for the calibrated model and 0.981 for the raw model. Bootstrap intervals on the 200-report test split indicate the uncertainty introduced by rare-class support: the calibrated accuracy was 0.855 (95% CI: 0.805–0.905), the macro-F1 was 0.826 (95% CI: 0.728–0.903), the class 2 recall was 0.700 (95% CI: 0.375–1.000), and the ECE was 0.069 (95% CI: 0.055–0.121).

Formal decision-curve analysis requires a municipal action threshold that encodes the relative cost of missed urgent cases and false alarms. No staff utility threshold, SLA threshold, or staffing-cost record was available in the public data. A workload simulation was therefore used as a reproducible open-data analogue for practical decision-support value. This is consistent with decision-curve analysis, which requires an action threshold or utility trade-off before a prediction score can be translated into net benefit [23]. Table 6 reports the workload simulation on the gold-standard test split and compares short-list review with full manual review.

Table 6. Workload simulation on the gold-standard test split. The table compares short-list review with full manual review.

Review Rule	Manual Workload	High-Urgency Cases Found	Operational Reading
Review top 10 model-ranked reports	10/200 reports, or 5% of the labeled test split	9/10 high-urgency cases	Precision at 10 = 0.900. This gives strong short-list enrichment under the small test split.
Review top 20 model-ranked reports	20/200 reports, or 10% of the labeled test split	10/10 high-urgency cases	All high-urgency test cases were captured in the first 20 reports.
Manual review without ranking	200/200 reports, or 100% of the labeled test split	10/10 if every case is read	The ranking reduces the first-pass review burden, but it does not replace staff judgment.
Closed-case priority deciles	Retrospective corpus-level proxy	Top priority decile median closure about 9–10 days. Lowest decile about 59–60 days.	This is compatible with existing triage pressure, but it is association only.

The gold-standard split is small, but it shows the intended operational use of the score. The model is used to order cases for human review, not to replace staff triage.

The simulation supports a narrow practical claim: the model can help create a compact review queue for possible high-urgency cases. It does not prove municipal workflow improvement, since no prospective user study or staff-decision comparison was available.

A five-fold comparison across alternative text baselines on the 1000 labeled reports produced mean macro-F1 values of 0.771 for TF-IDF logistic regression, 0.800 for TF-IDF linear SVM, and 0.578 for TF-IDF multinomial naive Bayes. Frozen transformer embedding baselines did not outperform the linear TF-IDF alternatives under the same label constraint: distilbert-base-multilingual-cased embeddings reached a mean macro-F1 of 0.642 and a class 2 recall of 0.460, while XLM-R embeddings reached a mean macro-F1 of 0.700 and a class 2 recall of 0.600. Each fold contained only 10 high-urgency cases. These results support the use of a linear discriminative baseline for auditability, but they do not support claims of state-of-the-art classification.

Table 7 shows that topic modeling surfaced a small number of dominant operational themes. The largest topic reflected sidewalk and road-surface hazards, with a volume of 244,568 reports, followed by templated acknowledgment or routing messages, with a volume of 144,160 reports. Across topics, mean urgency probabilities were relatively low, around 0.02 to 0.09, consistent with the rarity of high-urgency labels. Hotspot scores differentiated themes that are both frequent and spatially concentrated. Topics linked to physical hazards and obstructions, for example, sidewalk damage, bulky waste, and fallen trees, tended to show elevated hotspot scores relative to administrative or templated text. Some high hotspot values were associated with noisy tokens, for example, line-break artifacts, which shows the need for continued text hygiene in production deployment.

Table 7. Topic, urgency, and hotspot interaction. The table reports the largest topics by volume, with abridged top words and average urgency probability.

Topic	Top Words (Abridged)	Total Volume	Avg Urgency Prob.	Ag Hotspot Score
14	sidewalk, dangerous, hole, damaged, broken, bike,	244,568	0.0838	0.8402
11	thank, hello, regrettably, (template artifacts)	144,160	0.0860	0.7538
15	tree, height, boards, cardboard,	37,047	0.0396	1.0183
17	furniture, chair, board, wooden,	25,911	0.0302	0.9318
0	operator, forwarded, request,	14,258	0.0376	0.7280
5	bag, white, blue, uncollected,	11,538	0.0373	0.7329
8	street, corner, dirty, lighting,	11,176	0.0803	0.5498
4	waste, construction, bin,	5788	0.0467	0.6191
19	non compliant, regulatory, parking,	2945	0.0223	0.8391
3	deposit, clandestine, illegal,	3231	0.0342	0.6109

The supplementary topic comparison on 50,000 sampled reports produced similar UMass coherence for NMF (−3.344) and LDA (−3.355), with NMF slightly higher on this diagnostic. BERTopic on 10,000 sampled reports produced 29 non-outlier topics and assigned 30.3% of reports to the outlier cluster. Its largest themes were coherent operational categories such as bags, waste, bike paths, lighting, sidewalk holes, trees, clandestine deposits, panels, mattresses, and furniture. The comparison supports the descriptive use of topic models in this corpus, while the artifact-heavy terms and sizable BERTopic outlier share show that topic labels should remain interpretive and subject to domain review rather than being treated as stable operational categories.

The z-score-based detector did not flag events above the predefined anomaly threshold during the analysis window, indicating that fluctuations in report volume and predicted urgency remained below the chosen detection rule. This result should be read as threshold-dependent: for the evaluated period and aggregation level, the clearest operational signals are persistent hotspots and chronic issue themes rather than abrupt z-score anomalies.

Event-detection sensitivity checks show that the absence of events under the main z-threshold = 2.5 and six-week window is not invariant to parameter choices. At z-threshold = 2.5, an eight-week window flagged 151 area-metric events, and a 12-week window flagged 571. At z-threshold = 2.0 and a six-week window, 850 events were flagged. This confirms that the main result is a conservative thresholded finding rather than evidence that no temporal disruptions occurred.

SHAP analyses for the high-urgency class emphasized terms that plausibly encode safety risks, including dangerous, danger, risk, accident, crossing, and references to vulnerable road users (e.g., cyclists and pedestrians), as seen in Table 8. Also, for complaint-oriented reports, salient terms included abandoned, bags, clandestine, dirty, illegal, and recurrent. The resulting term lists align with expected municipal semantics and provide an audit trail that can be reviewed by domain experts (see Figure 3).

Table 8. Explainability audit. Highest-weighted terms by class are shown in abridged form.

Class	Term	Weight
High urgency (class 2)	Dangerous	7.883
High urgency (class 2)	Danger	6.022
High urgency (class 2)	Risk	5.896
High urgency (class 2)	Accident	5.399
High urgency (class 2)	Crossing	4.306
High urgency (class 2)	Cars	3.734
High urgency (class 2)	Falling	3.500
High urgency (class 2)	Cyclists	2.707
High urgency (class 2)	Bike	2.450
High urgency (class 2)	Pedestrians	2.100
High urgency (class 2)	Damage	1.634
High urgency (class 2)	Holes	1.534
High urgency (class 2)	Pedestrian	1.506
High urgency (class 2)	Marking	1.430
High urgency (class 2)	Weeks	1.479
Complaint (class 1)	Abandoned	5.167
Complaint (class 1)	bags	4.322
Complaint (class 1)	Bag	4.216
Complaint (class 1)	Not	3.684
Complaint (class 1)	Clandestine	3.283
Complaint (class 1)	Dirty	3.115
Complaint (class 1)	Again	3.091
Complaint (class 1)	Deposit	2.931
Complaint (class 1)	deposits	2.925
Complaint (class 1)	depot	2.691
Complaint (class 1)	Always	2.440
Complaint (class 1)	Illegal	2.250
Complaint (class 1)	Garbage	2.222
Complaint (class 1)	Depots	2.146
Complaint (class 1)	non	2.146

Table 9 summarizes the supplementary robustness and comparison checks conducted with existing labeled and unlabeled open data, including alternative text baselines, transformer embeddings, ranking metrics, uncertainty estimation, topic-model comparison, event-sensitivity analysis, label validation, geographic-bias screening, and circularity checks.

These checks use existing labeled and unlabeled open data. The blind second annotation validates the existing gold file and the 200-report model-ranked high-urgency candidate set. No external dataset nor non-public municipal record is used.

The second-human annotation matched all 1000 gold-standard records. Overall agreement was 0.531. Cohen's kappa was 0.246, and quadratic weighted kappa was 0.331. Agreement was uneven by class. The same label was assigned in 170 of 605 original class 0 reports, 322 of 345 original class 1 reports, and 39 of 50 original class 2 reports. The second annotator assigned 55 reports to class 2, close to the original 50. Many originally neutral reports moved into the routine complaint class, which indicates that the main disagreement

concerned the boundary between neutral and routine complaint rather than the existence of safety-risk language.

Label reliability is therefore explicitly reported in this version. The remaining limitation is not the absence of inter-annotator agreement, but the absence of adjudicated multi-annotator labels and the small number of true high-urgency reports.

For the 200 model-ranked candidate reports, the second annotator assigned all cases to class 2, with high confidence in 195 cases and medium confidence in 5 cases. Translation issues were flagged in 12 cases, and no cases were marked unclear. Because these reports were selected by the model for high urgency, this result should be interpreted as selective rare-class enrichment rather than an unbiased estimate of model performance.

The expanded validation subset contains 1000 annotated reports. The annotator assigned 20 reports to class 0, 541 to class 1, and 439 to class 2. Confidence is high for 790 reports, medium for 197, and low for 13. Translation issues are flagged in 28 reports, and 101 reports are marked unclear. The label distribution is not a corpus prevalence estimate because two strata are intentionally enriched for urgent or borderline cases. Table 10 reports the label distribution in the expanded validation subset and separates the random corpus stratum from the enriched strata.

The random stratum contains 42 high-urgency reports out of 500. The enriched strata add many more high-urgency examples, but they should not be read as the natural corpus distribution.

The existing calibrated model was then checked against these new human labels. On the random 500-report subset, the hard three-class predictions reached an accuracy of 0.388 and a macro-F1 of 0.382. For class 2, precision was 0.938, recall was 0.357, and average precision was 0.701. This indicates that default class decisions were conservative on new random cases, while the urgency score still ranked many safety-risk cases near the top. Table 11 reports model performance on the expanded validation subset by stratum, avoiding a direct mixture of prevalence-oriented and enrichment-oriented evidence.

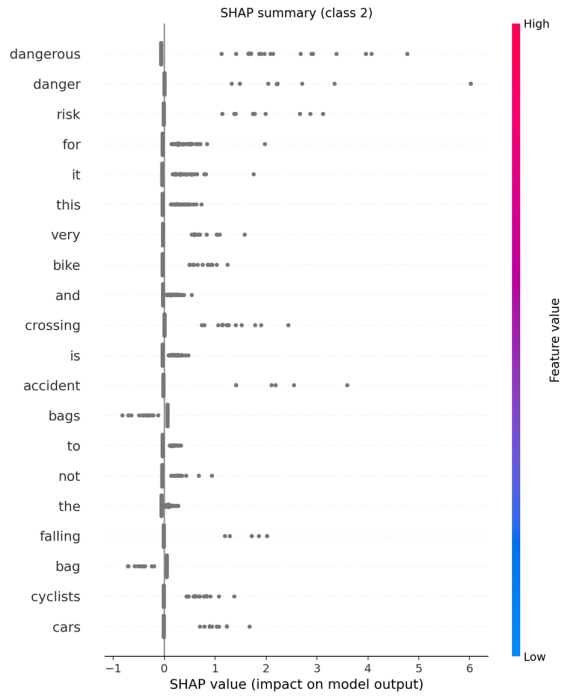


Figure 3. SHAP summary plot for the high-urgency class (class 2) in the TF–IDF logistic regression model. Terms such as dangerous, risk, and accident contribute positively to urgency predictions.

Table 9. Supplementary robustness and comparison checks using existing data.

Check	Data Used	Main Result	Interpretation
Classical text baselines	1000 labeled reports, five-fold CV	TF-IDF linear SVM macro-F1 = 0.800; TF-IDF logistic regression macro-F1 = 0.771; multinomial naive Bayes macro-F1 = 0.578.	Linear baselines remained competitive under the small gold-standard constraint.
Frozen transformer embeddings	1000 labeled reports, five-fold CV	DistilBERT multilingual macro-F1 = 0.642; XLM-R macro-F1 = 0.700.	Transformer embeddings did not outperform the transparent TF-IDF baselines without full fine-tuning.
Risk-aware ranking	200-report labeled test split	Calibrated top-10 recall for class 2 = 0.900; precision at 10 = 0.900; top-20 recall = 1.000.	Ranking metrics better matched prioritization than accuracy alone.
Uncertainty estimation	200-report labeled test split, 2000 bootstrap samples	Calibrated class 2 recall = 0.700, 95% CI: 0.375–1.000; ECE = 0.069, 95% CI: 0.055–0.121.	Rare-class results remained statistically fragile.
Topic-model comparison	Existing corpus samples	NMF UMass = −3.344; LDA UMass = −3.355; BERTopic produced 29 non-outlier topics with 30.3% outliers.	Topic outputs were useful descriptively but required domain review.
Event sensitivity	Existing area-week aggregates	At z = 2.5, eight-week window flagged 151 area-metric events; at z = 2.0, six-week window flagged 850.	The original no-event result was conservative and threshold-dependent.
Human second-pass label validation	1000 gold-standard reports, blind text-only second annotation	Overall agreement = 0.531; Cohen's kappa = 0.246; quadratic weighted kappa = 0.331; class 2 agreement = 39/50.	The high-urgency class showed stronger agreement than neutral labels, but the neutral/routine boundary remained unstable.
Descriptive geographic bias screen	522,132 open-data reports with postal-code fields	Mean predicted high-urgency probability varied from 0.028 to 0.139 across postal-code areas; text-presence rates varied from 0.543 to 0.941.	The checks identify variation requiring caution, but protected attributes and socioeconomic covariates were unavailable.
Operational and circularity boundary	Open-data status, timestamps, categories, and resolution-time proxies	Priority-decile and resolution-time patterns were retrospective associations rather than comparisons with staff decisions or deployment outcomes.	The model may reproduce existing administrative logic and cannot establish operational improvement without workflow access.
Circularity proxy model comparison	1000 labeled reports, five-fold CV	Text-only TF-IDF macro-F1 = 0.764; metadata-only macro-F1 = 0.579; category/department-only macro-F1 = 0.497.	Administrative metadata alone did not outperform text, but circularity remains possible because the labels and outcomes may still reflect platform logic.
High-urgency candidate review	200 model-ranked candidate reports from the existing corpus, blind human annotation	All 200 candidates were labeled class 2; 195 high-confidence and 5 medium-confidence labels; a total of 12 translation issues flagged.	This supports rare-class enrichment from existing data, but it remains model-selected and not an unbiased performance test.

Table 10. Label distribution in the expanded validation subset. The random stratum and enriched strata are reported separately because they answer different validation questions.

Annotation Stratum	N	Class 0	Class 1	Class 2	Class 2 Share
Random corpus reports	500	19	439	42	0.084
Model-ranked high-urgency candidates	300	0	4	296	0.987
Borderline or uncertain candidates	200	1	98	101	0.505
All expanded validation reports	1000	20	541	439	0.439

Table 11. Model performance on the expanded validation subset. Results are reported by stratum to avoid mixing prevalence-oriented and enrichment-oriented evidence.

Expanded Validation Model Check	N	Class 2 Support	Accuracy	Macro-F1	Class 2 Precision	Class 2 Recall	Class 2 Average Precision
Random corpus reports	500	42	0.388	0.382	0.938	0.357	0.701
Model-ranked high-urgency candidates	300	296	0.987	0.331	0.987	1.000	0.968
Borderline or uncertain candidates	200	101	0.240	0.180	0.545	0.059	0.616
All expanded validation reports	1000	439	0.538	0.470	0.969	0.722	0.940

The random stratum gives the most conservative external check. It shows high class 2 precision and modest class 2 recall under hard decisions, which supports ranking-based use more than autonomous classification.

Ranking metrics give a more operational reading. In the random 500-report subset, the top 10 reports ranked by predicted urgency contained 9 class 2 reports, and the top 20 contained 18 class 2 reports. Reviewing the top 100 captured 35 of the 42 class 2 reports. This supports the use of the score for short-listing cases for human review, not as an autonomous classifier. Table 12 reports the workload simulation for the random 500-report validation stratum and shows how many high-urgency cases are found when only the highest-ranked reports are reviewed.

Table 12. Workload simulation on the random 500-report validation stratum. The table reports how many high-urgency cases are found when only the highest-ranked reports are reviewed.

Random 500 Workload Rule	Reports Reviewed	Class 2 Found	Precision	Recall
Top 10 by urgency score	10	9/42	0.900	0.214
Top 20 by urgency score	20	18/42	0.900	0.429
Top 50 by urgency score	50	27/42	0.540	0.643
Top 100 by urgency score	100	35/42	0.350	0.833

This workload view is the clearest practical result from the expanded validation subset. Reviewing the top 100 cases captures most high-urgency reports in the random stratum, but the model still needs human review.

Descriptive bias screening using available postal-code fields showed material geographic variation in both model outputs and text availability. Across postal-code areas,

mean predicted high-urgency probability ranged from 0.028 to 0.139, while text-presence rates ranged from 0.543 to 0.941. In the labeled subset, high-urgency shares also varied by postal code, but several areas had very small labeled counts, so these differences were treated as instability signals rather than as evidence of socioeconomic bias. Because the open dataset does not include protected characteristics, household income, population exposure, complaint access rates, or internal service standards, the analysis cannot support a substantive fairness audit.

This fairness section is empirical but bounded. Earlier Brussels FixMyStreet research found socio-demographic participation differences across districts [24]. The current public export lacks the protected attributes, household income measures, population exposure denominators, access-rate denominators, and internal service standards needed to replicate a substantive fairness audit. The revised paper therefore reports what can be checked now: postal-code variation in predicted urgency, text availability, and labeled high-urgency shares. This limitation is consistent with 311 equity studies showing that resident-reported data can underrepresent some neighborhoods or service needs when reporting propensity differs from underlying conditions [4,5,25].

Aggregating closed cases by priority deciles revealed a non-linear relationship between predicted priority and median resolution time. The highest-priority decile showed the shortest median resolution time (approximately 9–10 days), while the lowest-priority decile exhibited the longest (approximately 59–60 days). Middle deciles clustered around 16–22 days, and the second-highest decile showed a longer median of approximately 35 days, consistent with a subset of high-salience but operationally complex cases.), consistent with a subset of high-salience but operationally complex cases. A complementary area-week scatter plot showed that very low priority scores coincide with a heavy-tailed distribution of resolution times, including extreme delays, whereas higher priority scores rarely coincide with long resolution times (see Figures 4 and 5).

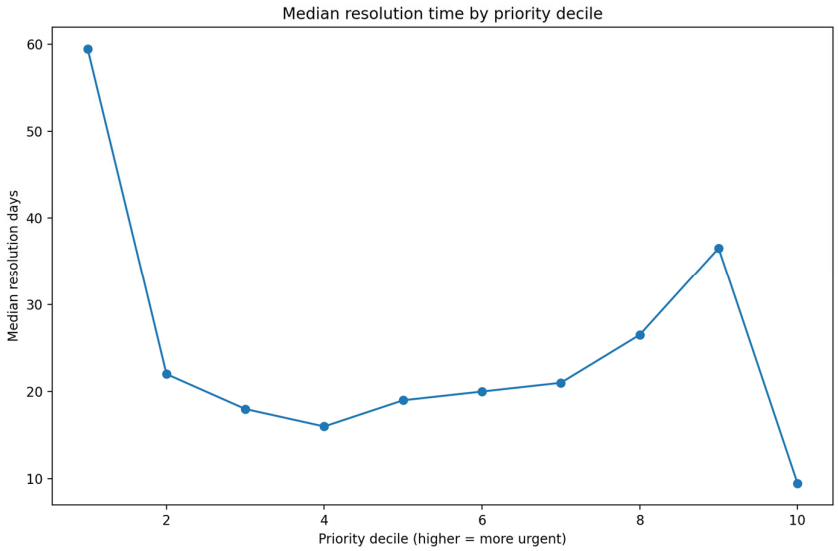


Figure 4. Median resolution time (days) by priority decile, where higher deciles correspond to higher predicted priority. The top decile is associated with the shortest median resolution time.

This pattern is compatible with existing triage processes, but it should be interpreted as an association rather than evidence that the model itself improves operational decisions.

The comparison with existing triage is limited to public workflow evidence. Official Brussels documentation describes FixMyStreet as a platform through which the public and authorities report incidents, monitor each step of incident resolution, and route broad issue types to actors such as Brussels Mobility, Bruxelles Proprete, and Sibelga [26]. The present

study can compare the model with visible platform fields, including category, responsible organization, status, postal code, and resolution time. It cannot compare against non-public staff decisions, dispatch rules, SLA files, or internal escalation notes.

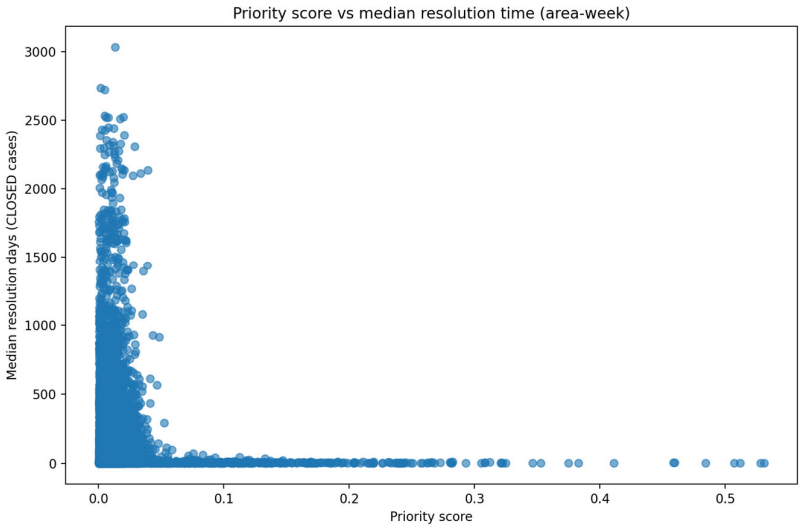


Figure 5. Priority score versus median resolution time aggregated at the area-week level (closed cases). Low priority scores exhibit a heavy-tailed distribution of resolution times.

5. Discussion

This study set out to translate high-volume, multilingual citizen reports into decision-support signals that remain auditable under municipal constraints. The results suggest that a transparent baseline can already deliver operational value when it is paired with probability calibration and explainability. This is consistent with the broader "citizens as sensors" framing of volunteered geographic information, where citizen input expands monitoring capacity but also introduces heterogeneity and bias that must be handled carefully [1,2]. The broader 311 literature supports this cautious stance: service requests can act as useful urban signals, but they also reflect the design of reporting channels and unequal participation patterns [3–7,25].

This study should be read as secondary analysis of publicly available civic-reporting data. This open-data scope makes the pipeline reproducible, but it also restricts what can be validated: the data do not contain annotator identities, adjudication histories, staff decisions, protected attributes, socioeconomic context, or records of how municipal officers actually used reports after submission.

Recent governance frameworks reinforce this cautious interpretation. The NIST AI Risk Management Framework emphasizes risk mapping, measurement, management, and monitoring for trustworthy AI systems, while the EU Artificial Intelligence Act places particular emphasis on fundamental-rights safeguards for high-risk AI uses [21,22]. The present pipeline is therefore best understood as an analytic aid for prioritization review, not as an autonomous allocation mechanism.

A central finding is the trade-off between discrimination and usable uncertainty. The raw TF–IDF + logistic regression model produced strong performance on the held-out gold split, and the confusion matrix indicates near-perfect separation for class 2 in that particular split (10/10 correct). However, the reliability analysis shows that the raw model's confidence is not uniformly trustworthy, especially at higher predicted probabilities. In public-service settings where scores may be interpreted as risk signals (not merely ranks), miscalibration can lead to overconfident triage and brittle decision rules. Post hoc calibration improved probability–accuracy alignment (ECE = 0.069) in the expected

direction [15], but it also shifted class decisions toward more conservative predictions, reducing class 2 recall on this split (7/10). Rather than presenting this as a net "win" or "loss," the evidence supports a more practical interpretation: calibration makes the score safer to treat as an estimated probability, while thresholding and workflow design must be tuned to municipal risk tolerance (e.g., prioritizing recall for safety hazards).

The topic and hotspot outputs support a second, complementary use case: moving from individual tickets to recurring themes and persistent spatial pressure. High-volume topics linked to sidewalks/road damage and bulky waste align with the kinds of chronic issues typically surfaced on reporting platforms. The fact that the z-score event detector did not flag shocks suggests that during the analyzed window, system dynamics were dominated by persistent patterns rather than abrupt anomalies. This is plausible for urban maintenance streams, where seasonality and backlog dominate variance and where threshold-based detectors intentionally avoid triggering on routine fluctuations. At the same time, the absence of detected events should be interpreted cautiously: aggregation choices (weekly bins and spatial units), the threshold level, and the smoothing implicit in "chronic" workflows can all reduce sensitivity to short-lived disruptions.

Explainability results provide governance-relevant face validity. SHAP attributions identified safety-related lexical cues (e.g., dangerous, risk, accident, and crossing) and complaint-oriented cues (e.g., abandoned, illegal, and dirty) as drivers of the corresponding class probabilities, which is consistent with the goal of producing reviewable "hooks" for domain experts [17]. Importantly, explainability also highlighted text hygiene issues: template artifacts (e.g., line-break tokens) appeared in topic terms, illustrating how operational systems can inject boilerplate that contaminates downstream models. This is not merely cosmetic; it can create spurious themes and inflate volume-based signals, which matters when prioritization is tied to hotspot scoring.

The resolution-time analysis suggests an association between predicted priority and operational throughput: high-priority deciles had shorter median resolution times, while low-priority deciles exhibited longer and more variable delays. This pattern is compatible with a functioning triage pipeline, but it should not be interpreted causally. Resolution time is affected by confounding factors such as task complexity, departmental capacity, and administrative routing, and text-derived priority could correlate with these latent variables rather than drive outcomes. The area-week scatter also indicates a heavy-tailed delay regime at low predicted priority, which may reflect backlog accumulation or hard-to-resolve categories rather than systematic neglect. From a decision-support perspective, the main takeaway is that priority scores can be used to structure queues and surface risk pockets, but they should be embedded in a policy-aware workflow (e.g., SLA rules and periodic review of low-priority backlogs) rather than used as an automated dispatcher.

Several limitations bound the conclusions and indicate clear next steps. The labeled evidence base is larger than the original gold standard, but it remains bounded. This study uses the original 1000-report gold standard, a 200-report model-ranked high-urgency candidate review, and a 1000-report expanded validation subset. The expanded validation subset contains 439 class 2 labels, including 42 class 2 reports in a random 500-report subset. This directly reduces the small-labeled-dataset concern, but it does not remove uncertainty. The enriched validation strata are not prevalence-representative, and all expanded labels come from one second annotator rather than from adjudicated multi-annotator review. The random subset also shows weaker three-class generalization than the original split, with macro-F1 of 0.382 and class 2 recall of 0.357 under hard model decisions. The stronger result is ranking-oriented: the top 100 random-subset reports captured 35 of 42 class 2 cases. The current results should therefore be read as stronger evidence for human-review short-listing than for autonomous triage.

Second, English normalization is a pragmatic choice for multilingual modeling, but translation and normalization can erase nuance (especially for short texts) and can introduce systematic artifacts. Comparative work on multilingual text categorization shows that translation-based pipelines and multilingual BERT-family models can differ materially in performance and transfer behavior [27]. In the supplementary benchmark, frozen XLM-R embeddings did not outperform the TF-IDF linear baselines on the current 1000-label gold standard. This result should not be interpreted as a general rejection of multilingual transformers, since full fine-tuning and larger rare-class support may change the comparison; it does show that transformer representations alone do not remove the present evaluation limitations.

The multilingual claim is also bounded by the preprocessing design. The model does not demonstrate native multilingual understanding; it uses English-normalized text derived from multilingual reports. The transformer embedding comparison and translation-related discussion show that multilingual representation choices matter, but they do not establish translation fidelity or cross-language equivalence. Stronger multilingual claims would require language-specific labels, source-language performance estimates, or independent assessment of translation quality, none of which were available in the public export. The second annotator flagged translation issues in 6 of the 1000 gold-standard records and 12 of the 200 additional candidate records, which reinforces the decision to treat multilingual validity as a limitation rather than as a settled result. Broader NLP evidence also cautions against treating multilingual systems as language-agnostic without language-specific evaluation, since resource asymmetries and translation quality can vary across languages [28,29].

Accordingly, this paper no longer makes a multilingual model-validity claim. The remaining claim is only that the pipeline analyzes English-normalized text derived from originally multilingual reports, with translation issues flagged as a limitation.

A further governance limitation concerns bias. Descriptive checks using existing fields show substantial variation in report volume, text presence, and predicted urgency across postal-code areas, but these patterns cannot be interpreted as socioeconomic bias without external covariates. The current data also do not support a full language-bias audit beyond translation-issue flags and text-source fields. These gaps matter because civic-reporting platforms reflect participation, access, administrative routing, and neighborhood visibility as much as underlying service need. The descriptive bias screen should be interpreted conservatively. Postal-code variation in text availability and predicted urgency may reflect unequal participation, different service needs, reporting habits, local infrastructure, or administrative routing. Without demographic covariates, exposure denominators, and internal service standards, the analysis can identify where bias may enter the pipeline, but it cannot estimate group fairness or disparate impact. Prior work on Boston, New York City, Brussels, and Kansas City shows that reporting volume can reflect access, propensity to complain, neighborhood resources, and resident-government interaction, not only objective service need [4,5,24,25].

Third, the hotspot score and event detector were designed for transparency, but they are not substitutes for formal spatial cluster inference. In particular, volume–urgency aggregation can confound reporting intensity with underlying incidence (e.g., participation biases across neighborhoods), a known concern in volunteered geographic information [2]. Careful deployment would therefore combine these signals with municipal context (e.g., population, footfall proxies, and inspection cycles) and would treat hotspots as triage leads, not ground truth.

Also, platform processes themselves shape the data. FixMyStreet-style systems include administrative routing, status transitions, and templated responses that affect both text and

outcomes [30,31]. Model monitoring should explicitly track changes in templates, category taxonomies, and departmental procedures because these can cause silent distribution shifts.

A final limitation concerns circularity between administrative labels, model targets, and operational outcomes. If the manually assigned labels were influenced by categories, status transitions, or priority concepts already embedded in the platform, the classifier may partly reproduce existing administrative logic rather than discover an independent notion of civic risk. The resolution-time association has the same limitation: faster closure for high-priority cases may reflect pre-existing triage practice, task simplicity, departmental capacity, or routing rules. Without direct access to municipal workflow records or a prospective comparison against staff decisions, the model should be interpreted as an auditable prioritization aid and diagnostic baseline, not as evidence of improved operational decision making. A proxy test directly compares text-only and metadata-only classifiers on the same 1000 labeled reports using five-fold stratified cross-validation. The text-only TF-IDF model reaches a macro-F1 = 0.764 and a class 2 recall = 0.720, while the metadata-only model using available category, department, status, postal-code, reporter, translation-issue, text-length, picture, and resolution fields reaches a macro-F1 = 0.579 and a class 2 recall = 0.620. A narrower category/department-only model reaches macro-F1 = 0.497. These results do not eliminate circularity risk, but they indicate that available administrative metadata alone do not reproduce text model performance.

The same human annotation protocol covers the 200 model-ranked high-urgency candidates drawn from unlabeled open-data reports. The annotator labels all 200 cases as class 2, with 195 high-confidence and 5 medium-confidence judgments. Translation issues are flagged in 12 records, and no cases are marked unclear. Because the queue is model-selected, this result should be interpreted as selective rare-class validation and as a reproducible route for extending rare-class annotation, not as an unbiased estimate of corpus-level performance.

6. Conclusions

This paper presented an explainable analytics pipeline with probability calibration for English-normalized FixMyStreet Brussels reports that integrates urgency modeling, topic discovery, and spatio-temporal hotspot scoring. On a manually annotated gold standard, a TF-IDF logistic regression baseline achieved reasonable classification performance on the labeled split, and post hoc calibration improved the trustworthiness of probability estimates at the cost of more conservative high-urgency predictions. Topic-level summaries and hotspot metrics highlighted dominant operational themes, including sidewalk and road hazards and bulky waste, while threshold-based event detection did not identify events above the predefined z-score threshold during the analyzed window. This result suggests that the observed dynamics are more visible as chronic patterns than as abrupt anomalies under the selected aggregation and threshold settings. Explainability audits provided linguistically intuitive drivers for urgent and complaint-oriented cases and surfaced data-quality artifacts that matter for governance.

Overall, the results support a cautious conclusion: simple, auditable models can provide preliminary triage structure when probabilities are treated as calibrated risk estimates, explanations remain available for review, and data-generation biases inherent to citizen reporting are explicitly acknowledged. The expanded annotation evidence base covers 2200 distinct human-reviewed reports and provides substantial rare-class evidence. It also shows that three-class generalization remains fragile on new random cases. Before these signals are used for resource allocation, future work should validate them against external operational indicators, strengthen source-language validation, and assess rare high-urgency predictions with adjudicated multi-annotator labels. The contribution is therefore limited

to a reproducible open-data prototype with transparent uncertainty, not a deployed or externally validated municipal triage system.

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